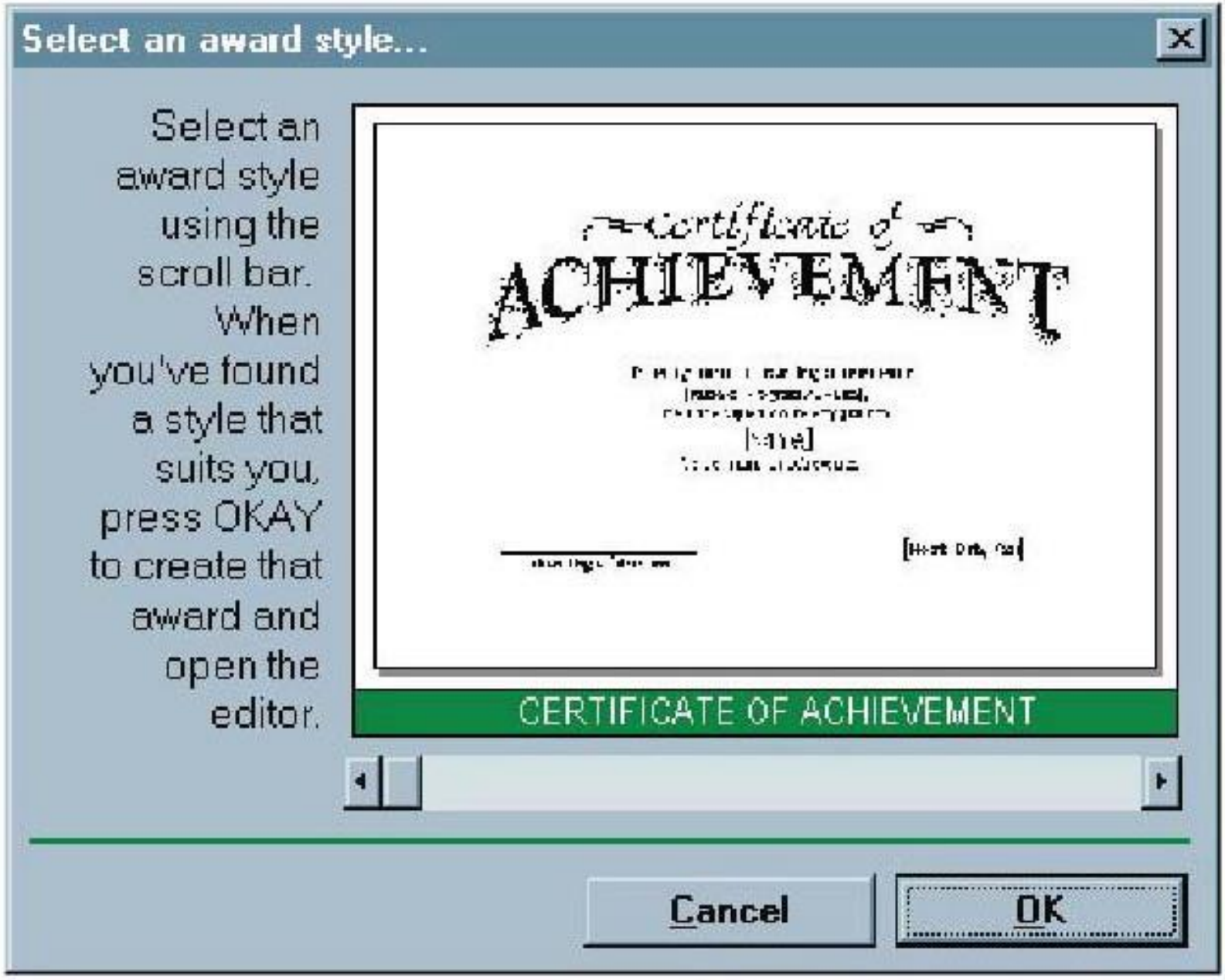
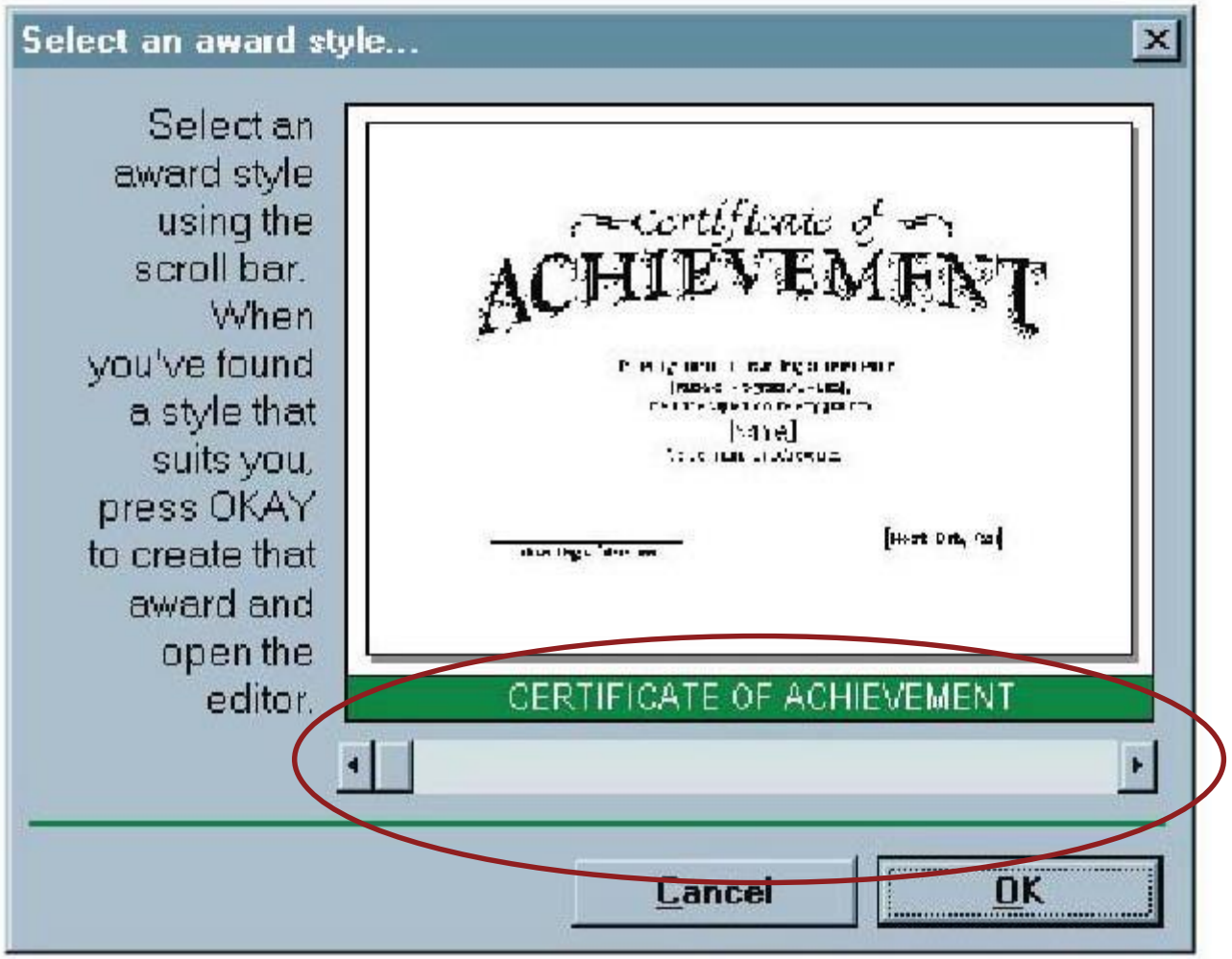
Usability

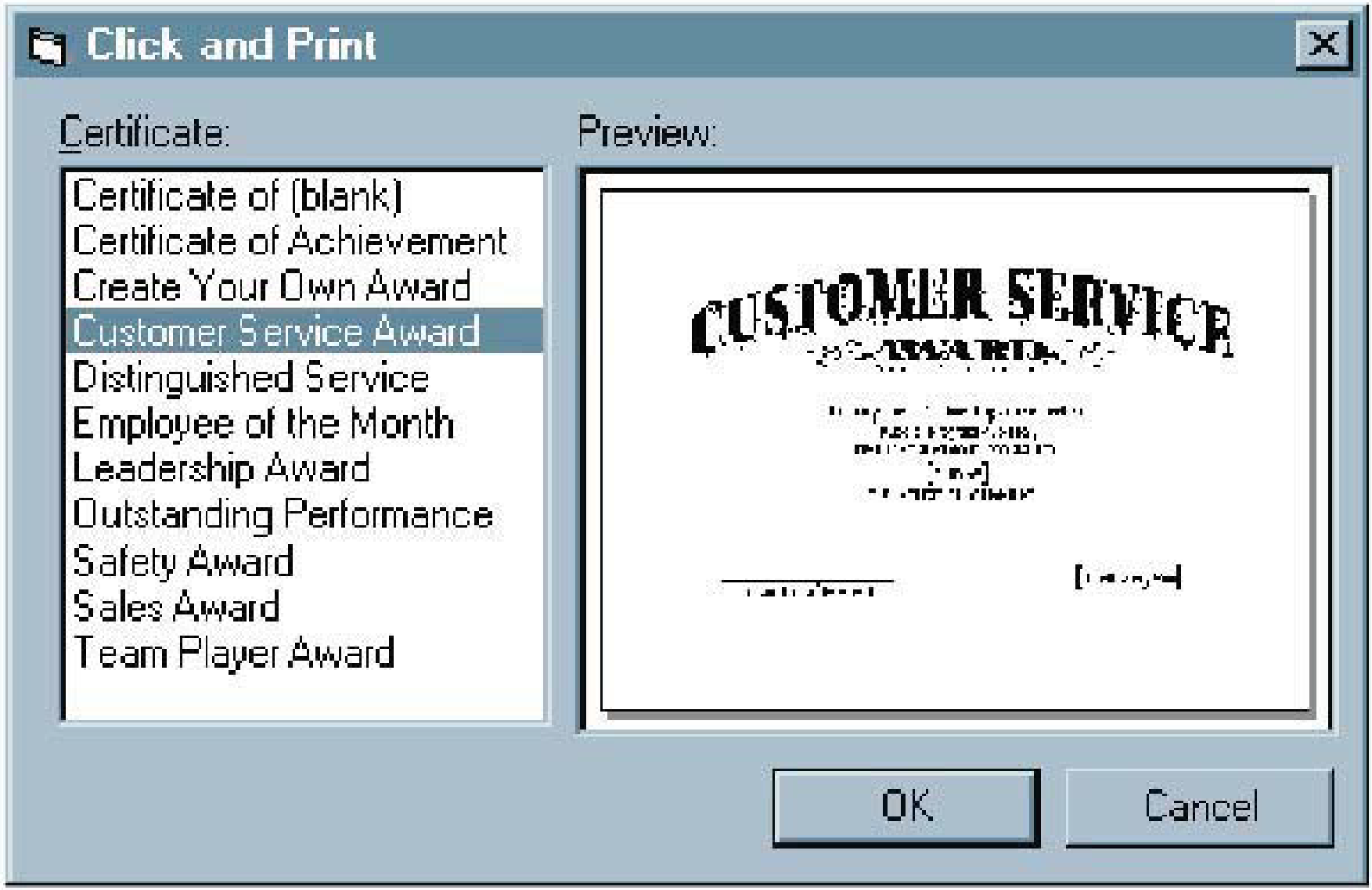

Select an award style...
Select an award style using the scroll bar. When you've found a style that suits you, press OKAY to create that award and open the editor.
Certificate of
ACHIEVEMENT
CERTIFICATE OF ACHIEVEMENT
Cancel
OK

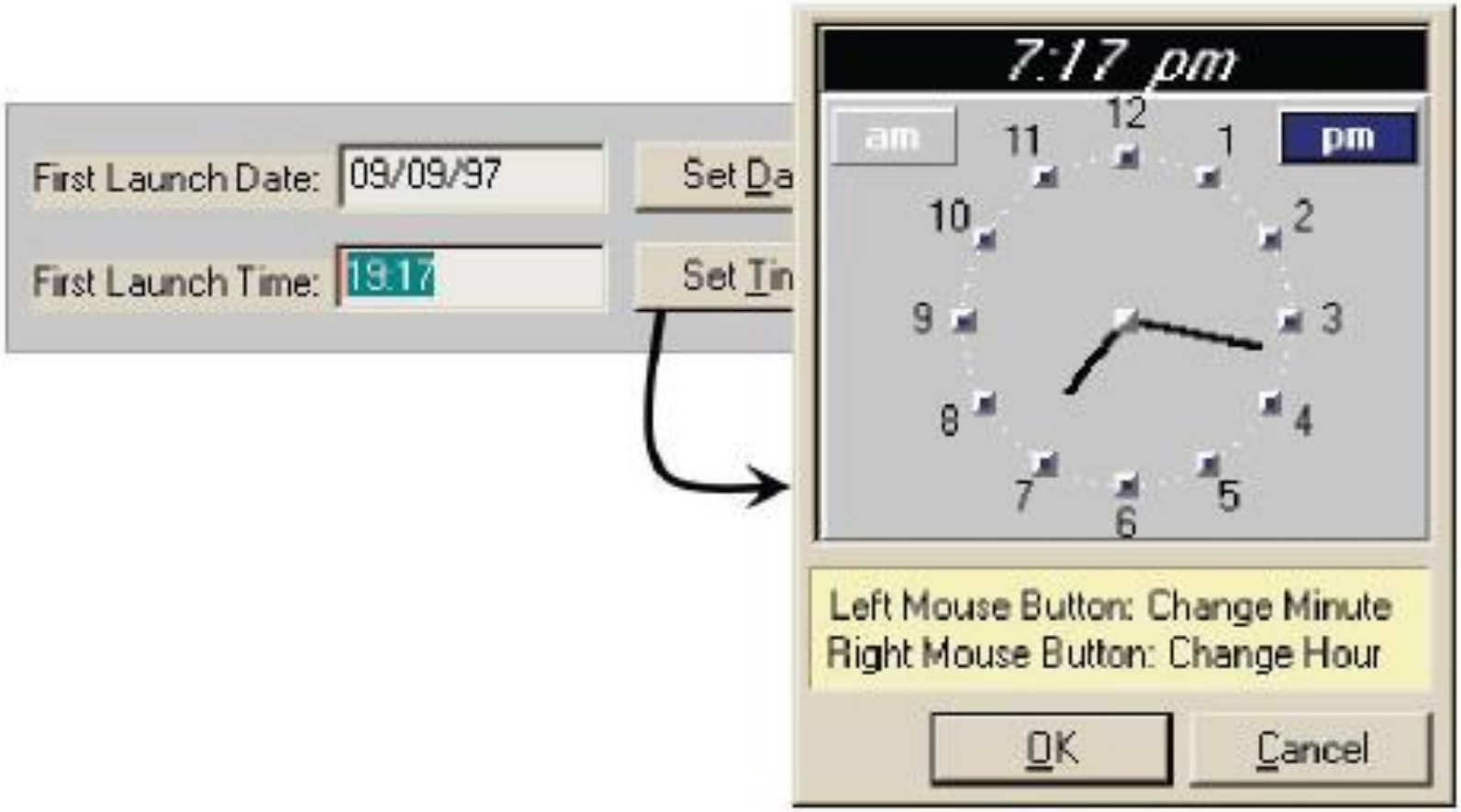
Why so much help for a simple task? "Press OKAY"? Where is that?

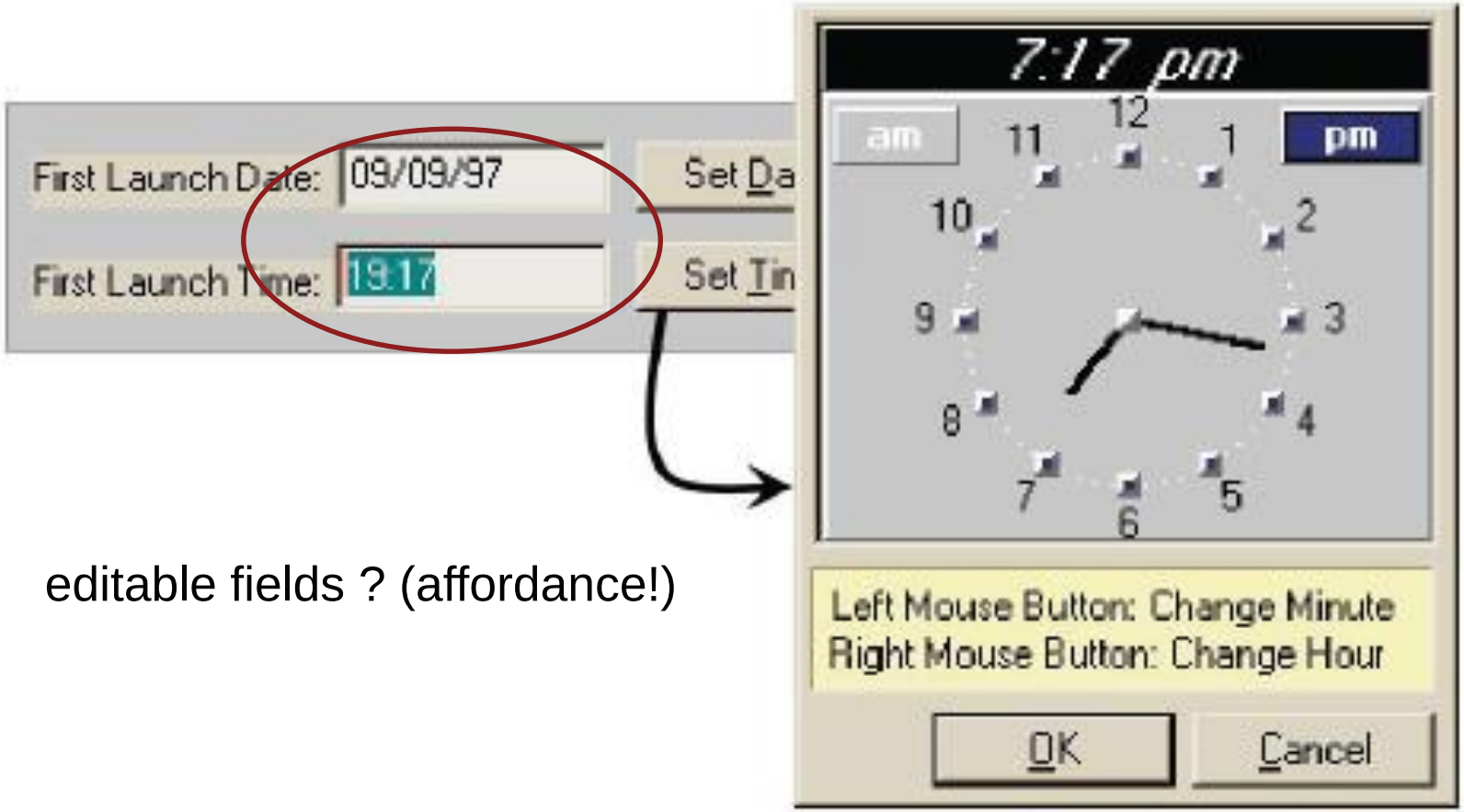


How many templates are there? How are they sorted? How far do you have to move the scrollbar to select the next one?

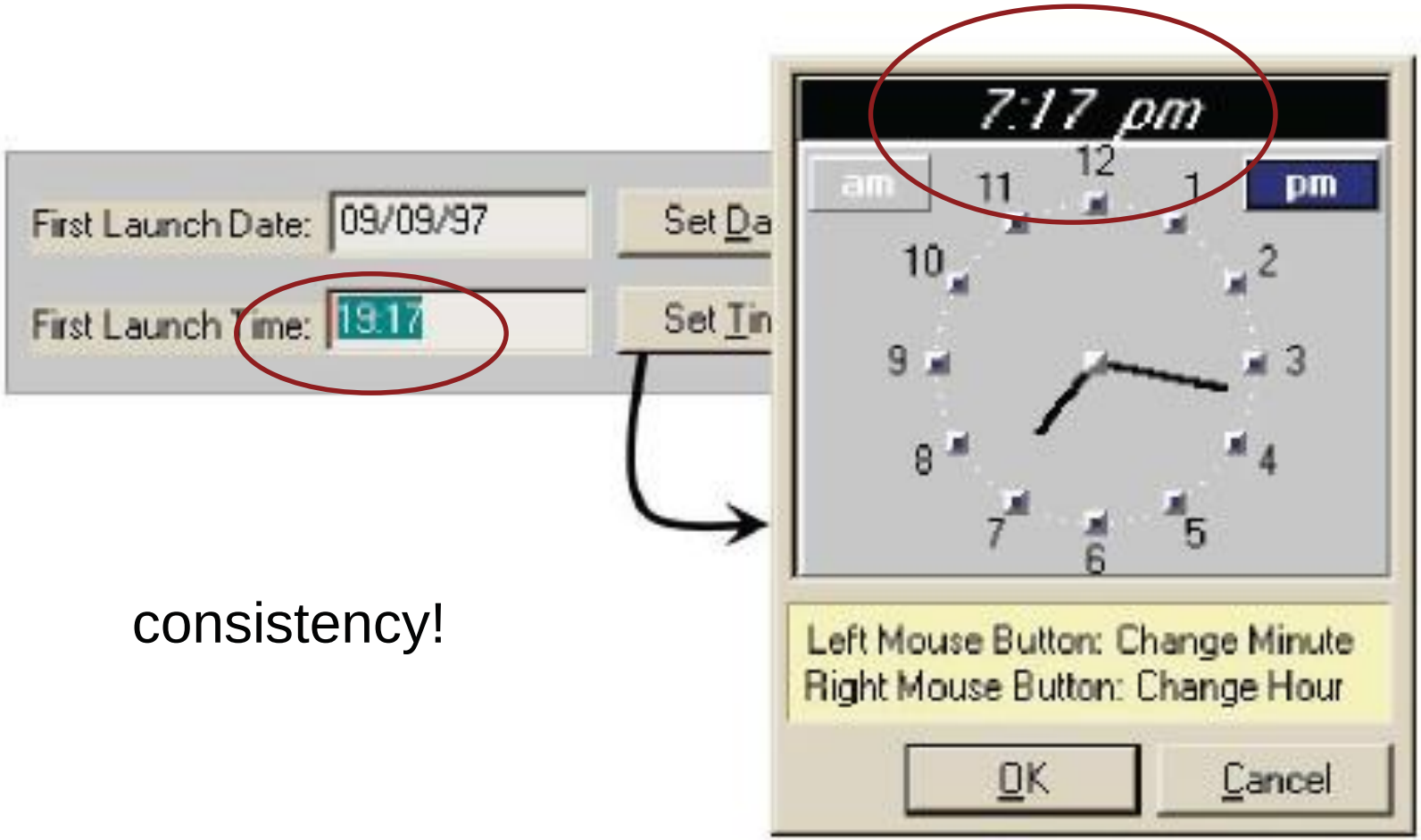
- A horizontal scrollbar underneath an image is designed for scrolling the content horizontally. **Inconsistency**
- Affordance for continuous scrolling, not for discrete selection
  - apparently-pullable door handles with a little sign that says "Push"
  - help text on this dialog box is filling the same role
- If a frequent user wants a template they've used before, how can they find it? Surely they'll remember that it's 56% of the way along the scrollbar?
  - no shortcuts for frequent users
  - Even the help text has usability problems
- How could this dialog box be redesigned to solve some of these problems?


Click and Print
Certificate:
Preview:
Certificate of (blank)
Certificate of Achievement
Create Your Own Award
Customer Service Award
Distinguished Service
Employee of the Month
Leadership Award
Outstanding Performance
Safety Award
Sales Award
Team Player Award
CUSTOMER SERVICE
OK
Cancel

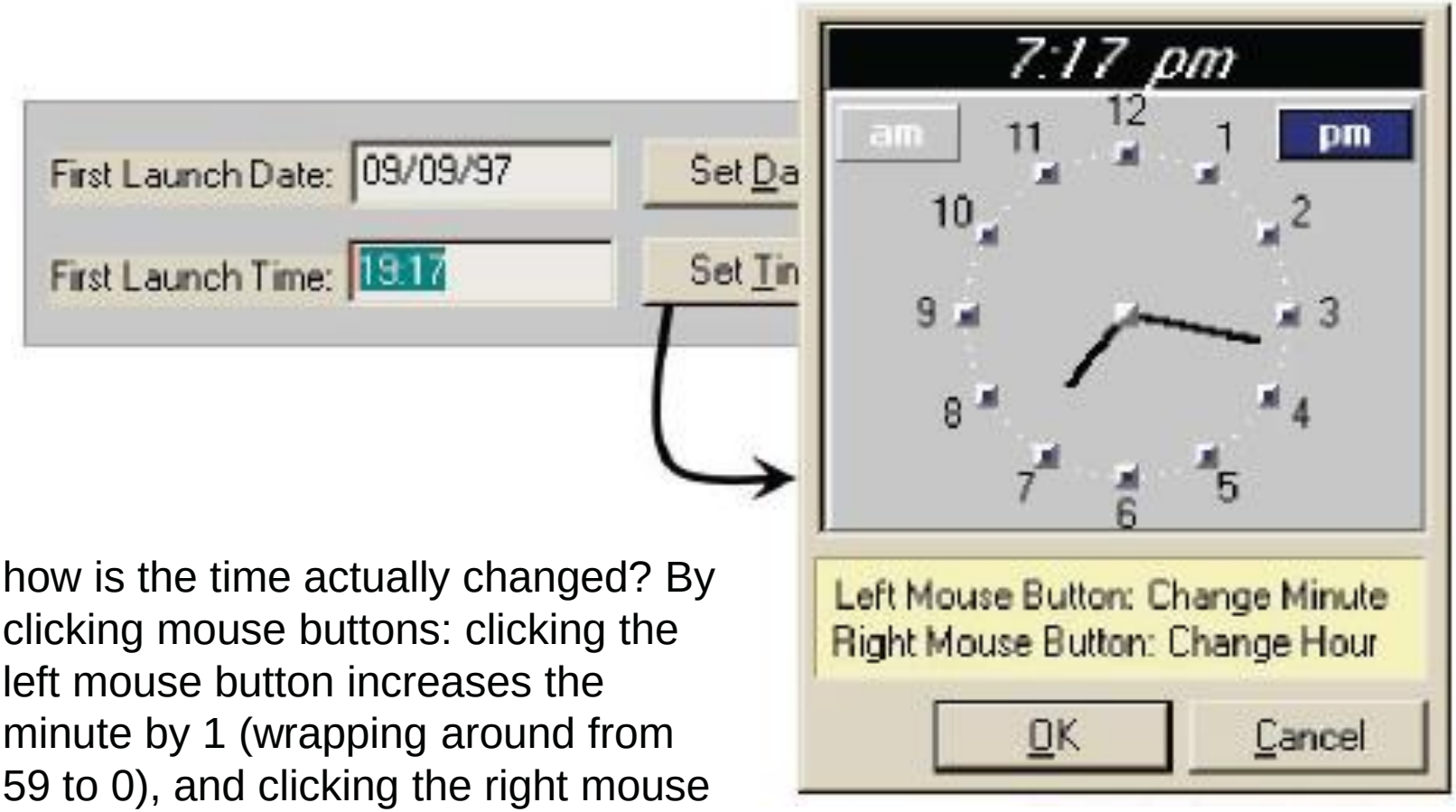

First Launch Date:
09/09/97
Set Da
First Launch Time:
19:17
Set Tin
7:17 pm
am
pm
12
1
2
3
4
5
6
7
8
9
10
11
Left Mouse Button: Change Minute
Right Mouse Button: Change Hour
OK
Cancel



editable fields ? (affordance!)



consistency!



how is the time actually changed? By clicking mouse buttons: clicking the left mouse button increases the minute by 1 (wrapping around from 59 to 0), and clicking the right mouse button increases the hour.

- designer has managed to turn a sophisticated graphical user interface, full of windows, buttons, widgets, and controlled by a hundred-key keyboard and two-button mouse, into a **clock radio!**
- not a result of laziness. Somebody went to a lot of effort to draw that clock face with hands. If only they'd spent some of that time thinking about usability instead

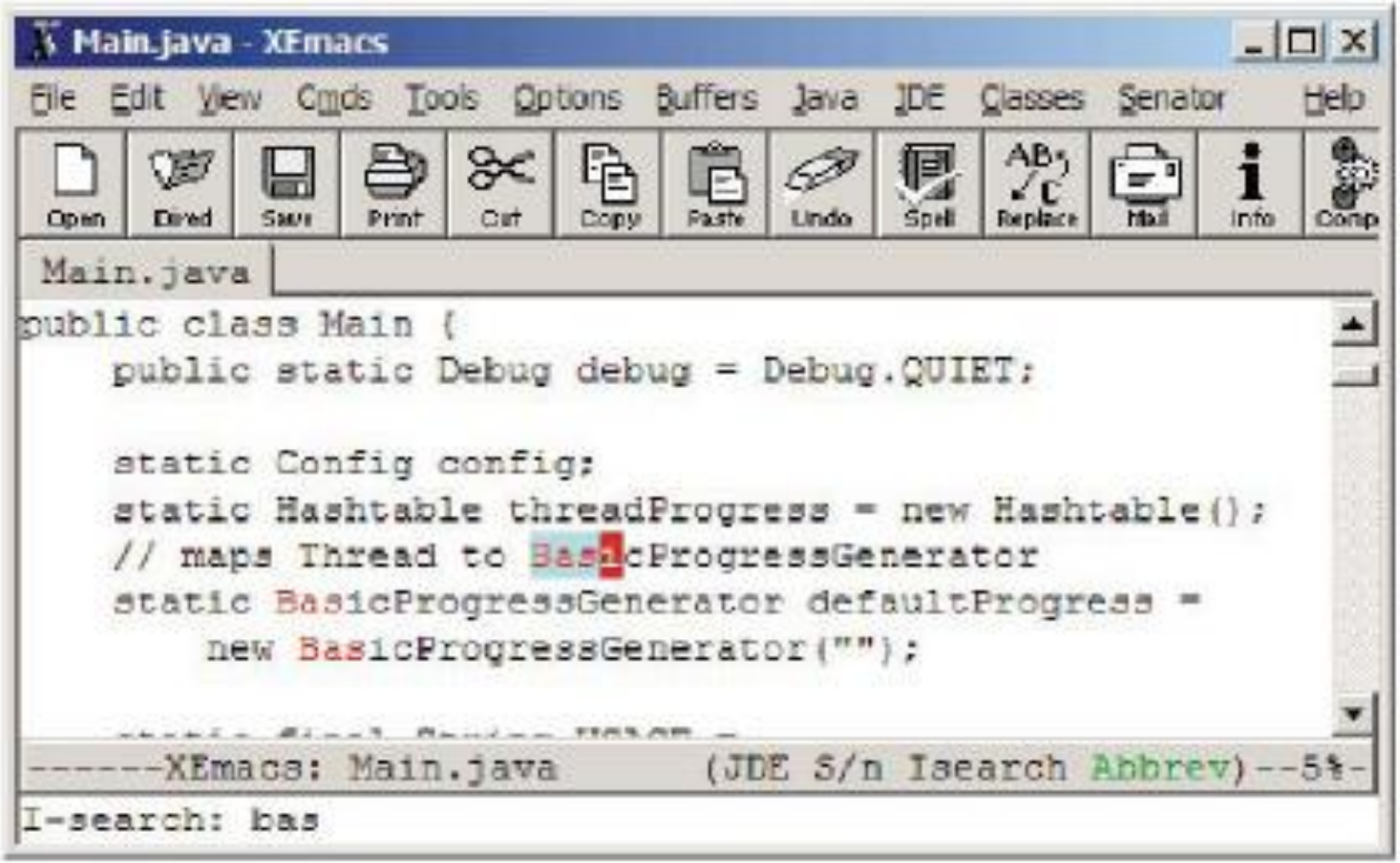
bridge-argenteuil.jpg-1.0 (RGB) 100%
File
Edit
Select
View
Image
Layers
Tools
Dialogs
Filters
Media
Video
Script-Fu
New... Ctl+N
Open... Ctl+O
Save Ctl+S
Save As...
Revert...
Page Setup
Print
Close Ctl+W
Quit Ctl+Q
bridge-argenteuil.jpg-1.0 (RGB) 100%

- no menu bar. Instead, all menus are accessed from a *context menu*, which pops up on right-click (inconsistency)
- faster to invoke, since you don't have to move the mouse up to the menu bar. A context menu can be popped up anywhere
- Really faster?

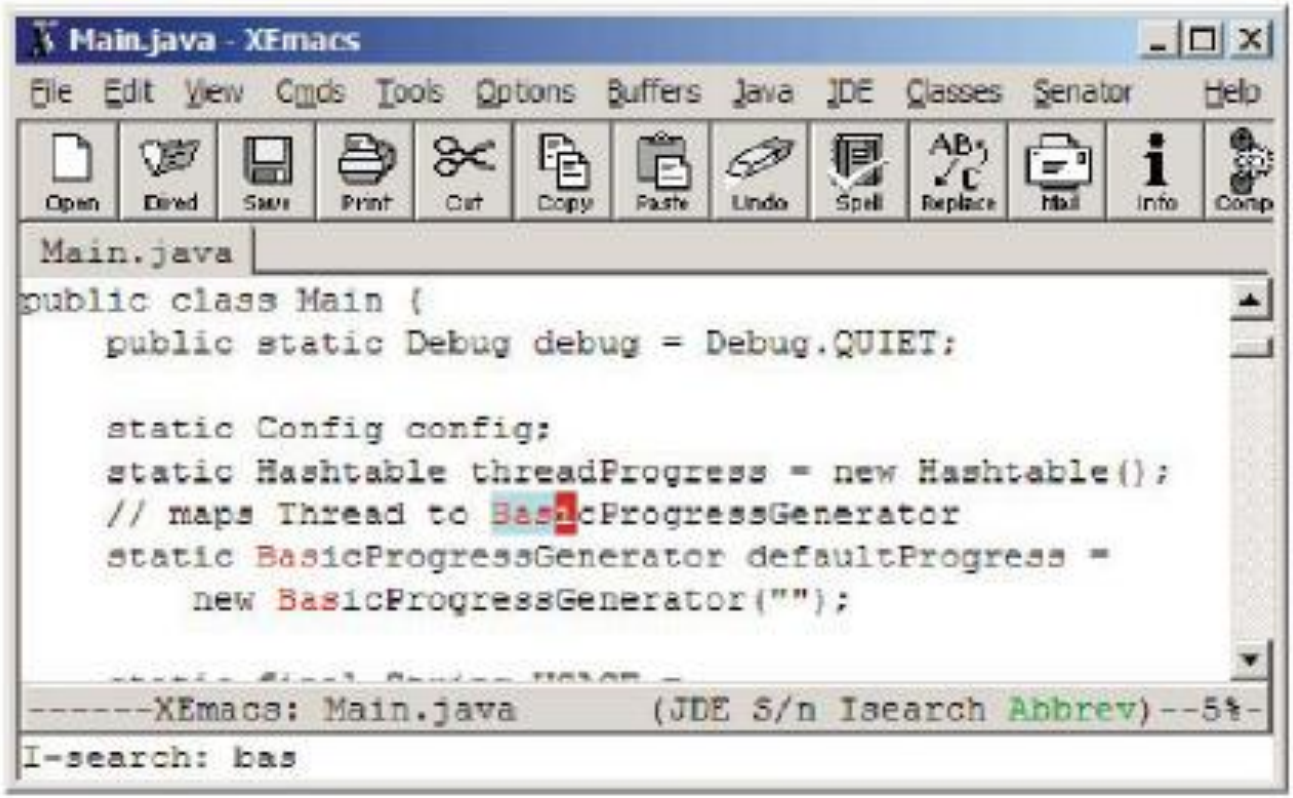
- if I want to reach an option on the File menu,
- I have to move my mouse carefully to the right, staying within the File choice, until it reaches the File submenu. If my mouse ever strays into the Edit item, the File menu I'm aiming for vanishes, replaced by the Edit menu.
- So if I want to select File/Quit, I can't just drag my mouse in a straight line from File to Quit – I have to drive into the File menu, turn 90 degrees and then drive down to Quit!

File
Edit
Select
View
Image
Layers
Tools
Dialogs
Filters
Media
Video
Script-Fu
New... Ctl+N
Open... Ctl+O
Save Ctl+S
Save As...
Revert...
Page Setup
Print
Close Ctl+W
Quit Ctl+Q
bridge-argenteuil.j
Cancel

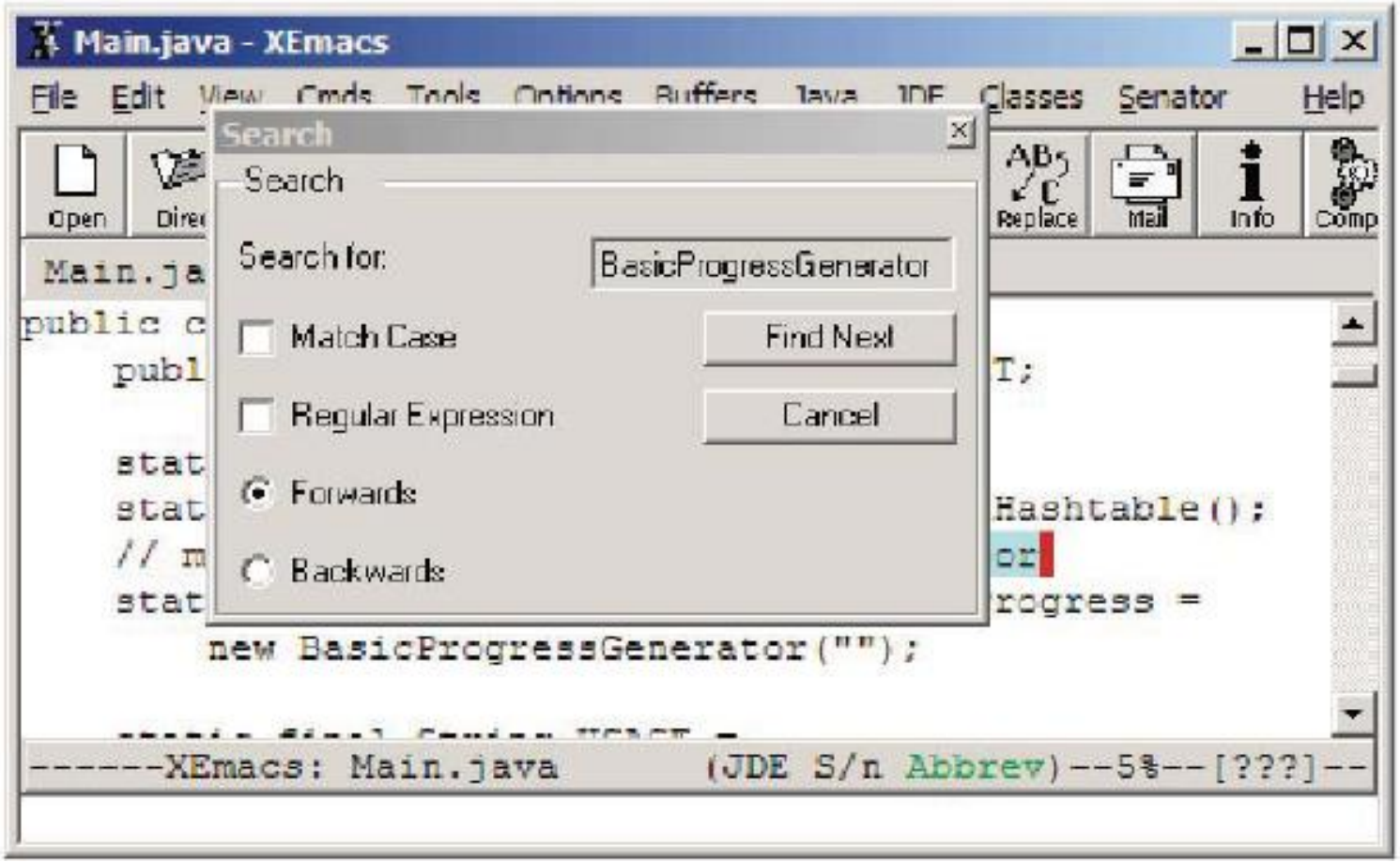
File
Edit
Select
View
Image
Layers
Tools
Dialogs
Filters
Media
Video
Script-Fu
New... Ctl+N
Open... Ctl+O
Save Ctl+O
Save As...
Revert...
Page Setup
Print
Close Ctl+W
Quit
bridge-argenteuil.j
Cancel



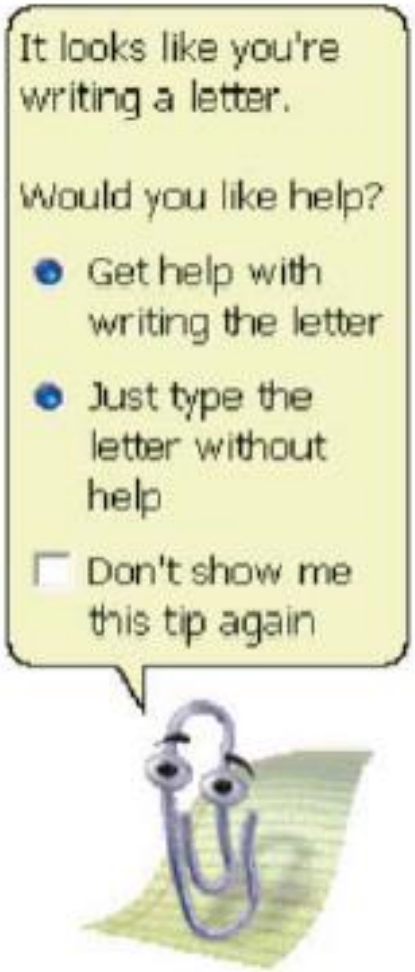
- Ctrl-S starts an *incremental search*
- only 3 letters was enough to find BasicProgressGenerator
- What's the downside?

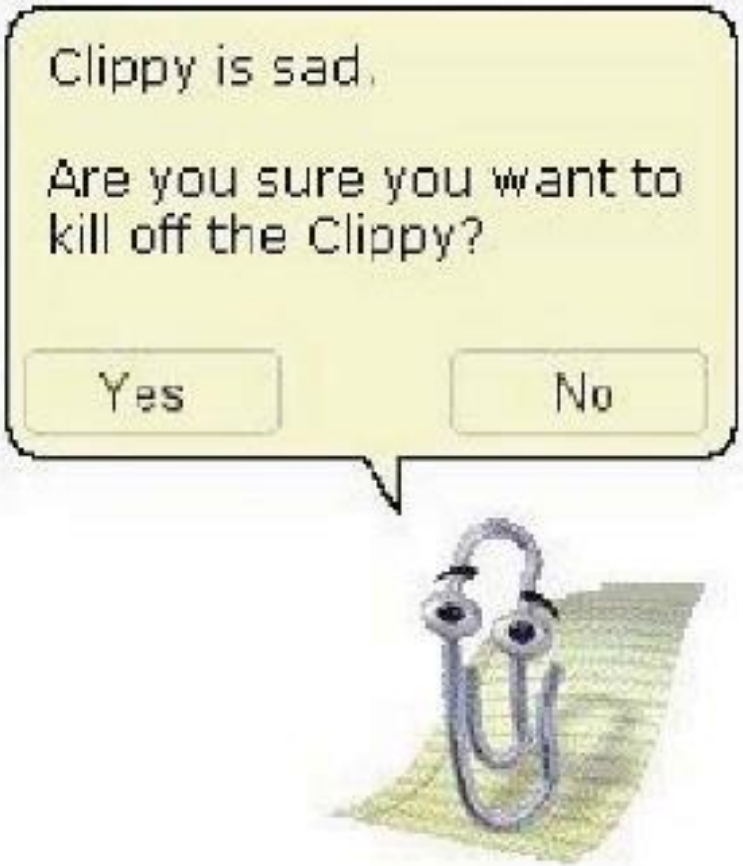


- All its controls are **invisible**.
- How do you start the incremental search? How do you search again? How do you go backwards? How do you do a case-sensitive search?



great on visibility,
but lacks the responsiveness, easy reversibility, and fast performance of incremental search
Even worse, it covers up the matches you're trying to find unless you manhandle it out of the way!





Users don’t read the manual, don’t use the online help, and don’t know how to find the answers to their problems

Clippy tries to suggest answers to the problem it thinks you’re having.

Unfortunately it’s often wrong, often intrusive, and often annoying.

- User interface strongly affects perception of software
  - Usable software sells better
  - Unusable web sites are abandoned
- Perception is sometimes superficial
  - Users blame themselves for UI failings
  - People who make buying decisions are not always end-users

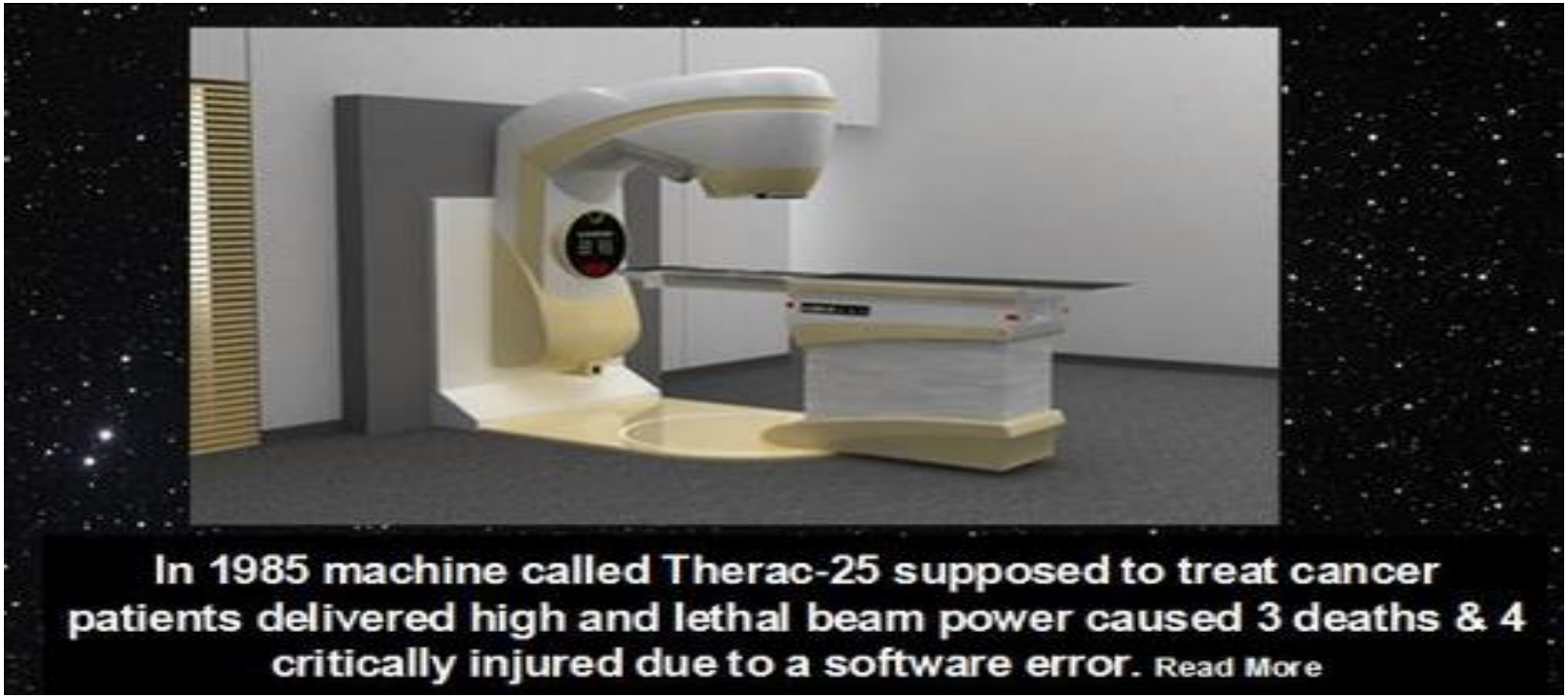
A poll found that ease of use (61%), customer service (58%), and no hassle installation (57%) are the most important factors US adults consider when purchasing a new technology product

If the program is slow, or crashes, or gets hacked, we know who to blame. If it's unusable, but not fatally so, the usability problems may go unreported

- Users' time isn't getting cheaper
- Design it correctly now, or pay for it later
- Disasters happen
  - Therac-25 radiation therapy machine
  - Aegis radar system in USS Vincennes
  - Supertanker accident off England
  - Predator UAV accident in Arizona

Users don't obey Moore's Law. Their time doesn't get cheaper with every new generation, like processors do. In fact, user time is probably getting *more* expensive every year. Interfaces that waste user time repeatedly impose a hidden cost that companies are less and less inclined to pay. For some applications, like customer call centers, saving a few seconds per call may translate into millions of dollars saved per year

For many software companies, a single customer support call can wipe out all the profit on that sale



two settings: a low-energy mode, beamed directly onto the patient, and a high-energy mode in which the beam was blocked by an X-ray generating filter.

system's design had a race condition between the user interface and the beam controller. If the operator chose a mode, and the machine started configuring itself, and then the operator backed up and made a different choice within the 8-second interval it took for the machine to swing its magnets, then part of the system wouldn't receive the new setting a fast, experienced operator could inadvertently give severe overdoses, and several patients died

In 1988, USS Vincennes guided missile cruiser shot down an Iranian airliner over the Persian Gulf with almost 300 people aboard

The radar operator interpreted the airliner as an F-14, descending as if to attack, rather than (in reality) a civilian plane that was climbing after takeoff

System was reporting the signal from an F14 on the ground at an airport hundreds of miles away, not the signal from the airliner; and the plane's altitude readout showed only its current altitude, not the direction of change in altitude

In April 2006, an Predator unmanned aerial vehicle (UAV) crashed while patrolling the US border in Arizona.
The pilot reported that his primary control console “locked up”, leading him to switch over to a backup console – but before switching between the two consoles, it’s necessary for the user to match the control positions on the new console to those on the original console.
The backup console was set to the fuel cutoff position (NO FUEL) when he did the switch over. As a result, the fuel was cut off to the UAV, and it went right down.

- You are not the user
  - Most software engineering is about communicating with other programmers
  - UI is about communicating with users
- The user is always right
  - Consistent problems are the system's fault
- ...but the user is not always right, either
  - Users aren't designers

In a study in 1950s, people were asked whether they would prefer lighter telephone handsets, on average they said they were happy with the heavy handsets they had

an actual test of telephone handsets, identical except for weight, revealed that people preferred the handsets that were about half the weight that was normal at the time.

Users don't always know what they want

- Google has discovered that when they survey users about how many search results they want per page (10, 20, 30), they say "30 results".
- when Google actually *deploys* 30-result search pages, usage drops by 20% relative to the conventional 10-result page.
- Probably because the 30-result page takes a half second longer to load

- Usability: how well users can use the system's functionality
- Dimensions of usability
  - Learnability: is it easy to learn?
  - Efficiency: once learned, is it fast to use?
  - Visibility: is the state of the system visible?
  - Errors: are errors few and recoverable?
  - Satisfaction: is it enjoyable to use?

how well users can use the system's functionality

Just as we can say algorithm X is faster than algorithm Y on some workload, we can say that interface X is more learnable, or more efficient, or less error prone than interface Y for some set of tasks and some class of users, by designing an experiment that measures the two interfaces

- Depends on the user
  - Novice users need learnability
  - Infrequent users need memorability
  - Experts need efficiency
- But no user is uniformly novice or expert
  - Domain experience
  - Application experience
  - Feature experience

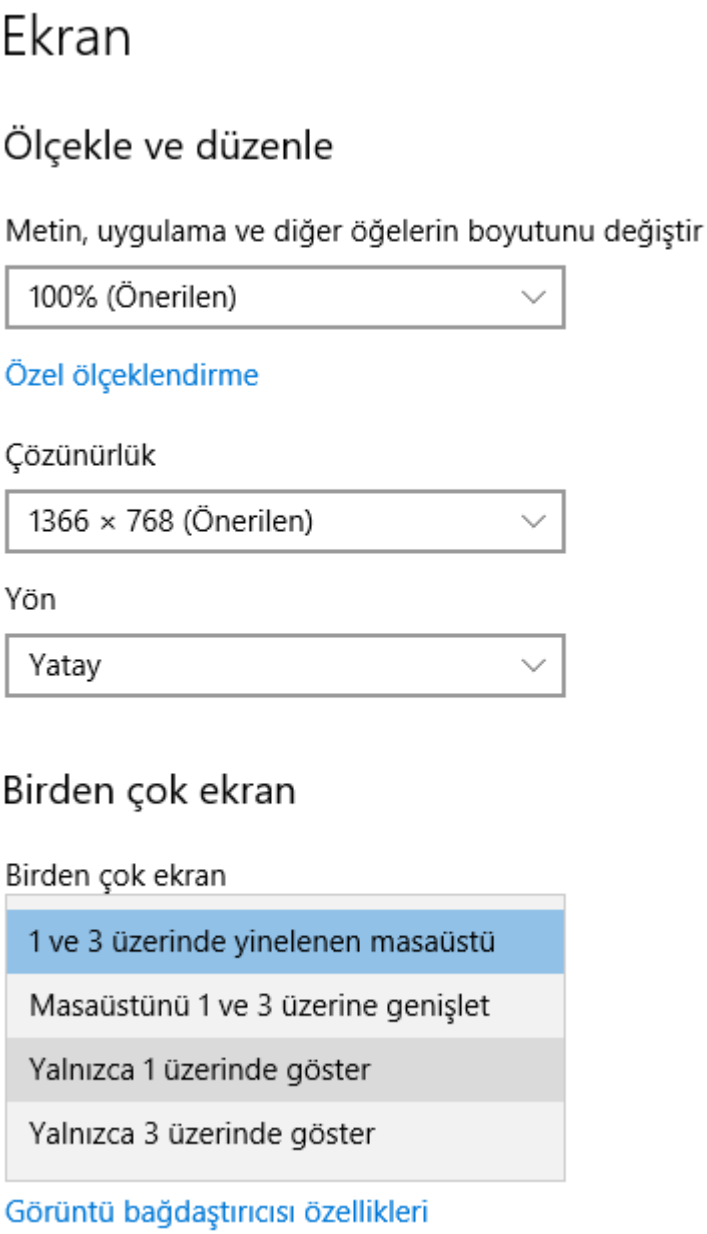
users may be *domain* experts, deeply knowledgeable about the stock market, and yet still be novices at your particular application

- Software designers have a lot to worry about:
  - Functionality
  - Performance
  - Cost
  - Security
  - **Usability**
  - Size
  - Reliability
  - Standards
- Many design decisions involve tradeoffs among different attributes
- We'll take an extreme position in this class

Astronauts may care more about reliability of their navigation computer than its usability; military systems would rather be secure than easy to log into

Short Story

- How I solved the projector problem



Ekran

Ölçekle ve düzenle

Metin, uygulama ve diğer öğelerin boyutunu değiştir

100% (Önerilen)

Özel ölçeklendirme

Çözünürlük

1366 × 768 (Önerilen)

Yön

Yatay

Birden çok ekran

Birden çok ekran

- 1 ve 4 üzerinde yinelenen masaüstü
- Masaüstünü 1 ve 4 üzerine genişlet
- Yalnızca 1 üzerinde göster
- Yalnızca 4 üzerinde göster

Görüntü bağdaştırıcısı özellikleri

- Repeat for «1 and 5»

Ekran

Ölçekle ve düzenle

Metin, uygulama ve diğer öğelerin boyutunu değiştir

100% (Önerilen)

Özel ölçeklendirme

Çözünürlük

1366 × 768 (Önerilen)

Yön

1 ve 2 üzerinde yinelenen masaüstü

1 ve 3 üzerinde yinelenen masaüstü

1 ve 4 üzerinde yinelenen masaüstü

1 ve 5 üzerinde yinelenen masaüstü

Masaüstünü bu ekrana genişlet

Bunu ana ekranım yap

Kablosuz bir ekrana bağlan

Görüntü bağdaştırıcısı özellikleri

Ekran

Ölçekle ve düzenle

Metin, uygulama ve diğer öğelerin boyutunu değiştir

100% (Önerilen)

Özel ölçeklendirme

Çözünürlük

1366 × 768 (Önerilen)

Yön

Yatay

Birden çok ekran

Birden çok ekran

1 ve 2 üzerinde yinelenen masaüstü

☑ Bunu ana ekranım yap

Kablosuz bir ekrana bağlan

Görüntü bağdaştırıcısı özellikleri

- You need to find out if the number indicate the possible index of a currently connected display device or a unique id given to a specific device